

SRF-PLL Based Active/Reactive Power Control, LCL Filtering, and SVPWM Analysis for a 25 kW Three-Phase Grid-Connected Inverter

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1. Project Purpose

This study examines the design and control of a 25 kW voltage-source inverter (VSI) supplied by a 700 V DC bus and connected to a 400 V (line-to-line), 50 Hz three-phase grid. The primary objective of the project is to establish a system architecture that ensures reliable phase synchronization with the grid, provides independent control of active and reactive power components, and performs low-harmonic current injection.

To achieve these goals, the system integrates SRF-PLL based synchronization, active and reactive power control within the dq reference frame, PI-based current regulation with feed-forward compensation, LCL filtering for high-frequency harmonic attenuation, and Space Vector Pulse Width Modulation (SVPWM) for optimized switching. The designed structure has been rigorously evaluated in a simulation environment regarding both power quality and control performance.

At the conclusion of the study, the inverter's success in tracking reference power values, the accuracy of the PLL synchronization, the quality of the current waveforms, and the harmonic suppression effectiveness of the LCL filter were successfully validated through simulation results.

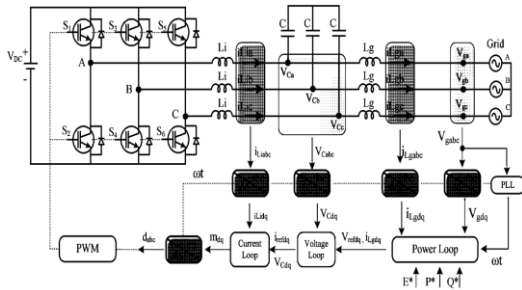


Figure 1. General control block diagram of the three-phase inverter system including power, voltage, and current control loops.

2. Reasons for Selecting the Three-Phase Grid-Connected Inverter Topology

In this study, a two-level voltage source inverter (VSI) topology, capable of transferring energy to a three-phase grid at a 25 kW power level, has been selected. The rationale behind this choice is based on criteria involving power capacity, circuit complexity, and controllability.

Transferring 25 kW to the grid via single-phase structures introduces severe current stress on semiconductor components and significant inherent power fluctuations. In contrast, the three-phase structure ensures a balanced power distribution among phases, providing a more stable energy transfer while facilitating the selection of semiconductors and filters by lowering per-phase current levels.

The primary factor in choosing the classic two-level structure over multi-level topologies is hardware and control simplicity. While multi-level inverters offer lower harmonic content, they necessitate an increased number of switches and complex gate-driving algorithms. In this work, the harmonic disadvantages of the two-level topology are mitigated by combining LCL filtering and SVPWM (Space Vector Pulse Width Modulation) techniques to achieve power quality compliant with grid standards.

Consequently, the two-level three-phase VSI topology was determined as the core power stage due to its industrial prevalence, direct compatibility with vector control algorithms in the dq frame, and limited hardware complexity.

3. Design Inputs and Main Technical Specifications

The fundamental electrical parameters and control inputs used in the design of the grid-connected inverter system are presented in Table 1. These values serve as the basis for establishing the power stage, determining the filter components, configuring the synchronization architecture, and implementing the active and reactive power control.

Table 1. Design inputs and main technical specifications.

Parameter	Symbol	Value
DC Bus Voltage	V_{dc}	700 V
Grid Voltage (Line-to-Line, RMS)	V_{LL}	400 V
Grid Frequency	f_{grid}	50 Hz
Grid Angular Frequency	ω	314.16 rad/s
Active Power Reference	P_{ref}	25 kW
Reactive Power Reference	Q_{ref}	15 kVAr
Switching Frequency	f_s	20 kHz
Inverter Topology	-	Three-phase two-level VSI
Modulation Method	-	SVPWM
Synchronization Method	-	SRF-PLL
Filter Type	-	LCL
Inverter-side Inductance	L_1	1 mH
Grid-side Inductance	L_2	0.2 mH
Filter Capacitance	C_f	10 μ F
Damping Resistance	R_f	3 Ω
Simulation Environment	-	PSIM

Phase-to-neutral RMS voltage for three-phase grid

$$V_{ph} = \frac{V_{LL}}{\sqrt{3}}$$

...is calculated by the relation. Accordingly,

$$V_{ph} = \frac{400}{\sqrt{3}} \approx 230.94 \text{ V}$$

...is obtained. The peak value of the phase voltage is...

$$V_{ph,peak} = \sqrt{2} V_{ph}$$

...and its numerical value is...

$$V_{ph,peak} = \sqrt{2} \times 230.94 \approx 326.6 \text{ V}$$

is found as shown. The $V_d \approx 326.6$ V value obtained in the simulation demonstrates that the synchronous reference frame has been accurately established. Based on the reference active and reactive power values, the apparent power of the system

$$S = \sqrt{P^2 + Q^2}$$

is calculated by the expression. Accordingly,

$$S = \sqrt{25^2 + 15^2} \approx 29.15 \text{ kVA}$$

is found. The power factor is

$$PF = \frac{P}{S}$$

from the relation

$$PF = \frac{25}{29.15} \approx 0.857$$

is obtained as. This result indicates that the system transfers not only active power but also a certain level of reactive power. The resonance frequency of the LCL filter is approximately

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{L_1 + L_2}{L_1 L_2 C_f}}$$

is determined by the expression. Numerically

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{0.001 + 0.0002}{0.001 \cdot 0.0002 \cdot 10 \times 10^{-6}}} \approx 3.1 \text{ kHz}$$

is obtained. This value indicates that the resonance frequency is positioned significantly above the grid frequency and below the switching frequency. Thus, while the filter has a limited effect on the fundamental component, it effectively suppresses switching harmonics.

Based on these technical data, the system was modeled to achieve the transfer of 25 kW active power and 15 kVAr reactive power; it was analyzed in the PSIM environment using SRF-PLL, power/current control in the dq frame, and an SVPWM-based modulation structure.

4. System Architecture and General Operating Principle

The designed system consists of a three-phase two-level voltage source inverter, an LCL filter, a grid measurement unit, an SRF-PLL-based synchronization block, power and current control loops in the dq frame, and an SVPWM modulation unit. The power stage generates a three-phase AC voltage by switching the 700 V DC bus voltage, which is then applied to the grid through the LCL filter.

In the control structure, three-phase grid voltages are first measured, and the instantaneous phase angle of the system is obtained via the SRF-PLL block. Using this angle information, the voltage and current magnitudes are transformed into the dq reference frame. Consequently, the sinusoidal three-phase quantities are expressed as stationary or slowly varying components, which are more suitable for control purposes.

In the outer control loop, the d- and q-axis current references are generated using the active and reactive power references. In the inner control loop, these current references are compared with the measured current components, and the voltage references for the inverter in the dq frame are obtained using PI-based controllers.

The resulting voltage references are transferred back to the stationary frame through an inverse transformation and applied to the SVPWM modulation block. The SVPWM algorithm determines the switching states corresponding to these references and generates the gate drive signals for the inverter's six switches. Thus, the DC bus energy is transferred to the grid in a controlled and synchronized manner.

The LCL filter incorporated into the system architecture suppresses high-frequency harmonic components arising from the inverter switching, thereby enhancing the power quality of the current injected into the grid. Therefore, the system is treated as an integrated structure that fulfills not only power transfer but also synchronization, current regulation, modulation, and harmonic suppression functions simultaneously.

5. Three-Phase Inverter and LCL Filter Design

In this section, the compatibility between the grid voltage level and the DC bus voltage, the current levels corresponding to the targeted power transfer, and the fundamental design indicators of the LCL filter are presented.

Since the line-to-line RMS voltage in a three-phase grid is $V_{LL} = 400V$, the phase-to-neutral RMS voltage is:

$$V_{ph} = \frac{V_{LL}}{\sqrt{3}}$$

is calculated by the relation:

$$V_{ph} = \frac{400}{\sqrt{3}} \approx 230.94 V$$

The peak value of the phase voltage is:

$$V_{ph,peak} = \sqrt{2} V_{ph}$$

So;

$$V_{ph,peak} = \sqrt{2} \times 230.94 \approx 326.6 V$$

is found as shown. This value is taken as a reference for the inverter to be able to generate the required fundamental component voltage for the grid.

Under active $P_{ref} = 25 kW$ and reactive power references $Q_{ref} = 15 kVar$, The apparent power of the system is calculated as:

$$S = \sqrt{25^2 + 15^2} \approx 29.15 kVA$$

Yielding a power factor of:

$$PF = \frac{25}{29.15} \approx 0.857$$

In three-phase systems, the relationship between apparent power and line current is:

$$S = \sqrt{3} V_{LL} I_L$$

From this, the line current is written as:

$$I_L = \frac{S}{\sqrt{3} V_{LL}}$$

and is numerically found as:

$$I_L = \frac{29.15 \times 10^3}{\sqrt{3} \times 400} \approx 42.1 \text{ A}$$

This current level constitutes a fundamental design input for semiconductor selection and filter sizing.

The resonance frequency of the LCL filter placed between the inverter and the grid is approximately calculated by the relation:

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{L_1 + L_2}{L_1 L_2 C_f}}$$

In this study, for:

$$L_1 = 1 \text{ mH}, L_2 = 0.2 \text{ mH} \text{ ve } C_f = 10 \text{ }\mu\text{F}$$

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{0.001 + 0.0002}{0.001 \cdot 0.0002 \cdot 10 \times 10^{-6}}} \approx 3.1 \text{ kHz}$$

is obtained. Selecting the resonance frequency to be significantly higher than the grid frequency and lower than the switching frequency aims to minimize the impact on the fundamental component while ensuring the effective suppression of switching harmonics. Additionally, a damping resistor of $R_f = 3 \text{ }\Omega$ was added to the filter to limit the magnitude peaks that may occur in the resonance region.

Based on these design parameters, the inverter-LCL filter-grid system was modeled, and verification was performed in the simulation section in terms of both power transfer and harmonic performance.

6. SRF-PLL Based Grid Synchronization

In grid-connected inverter applications, the control system must be in phase synchronization with the grid voltage to function correctly. For this purpose, the Synchronous Reference Frame Phase-Locked Loop (SRF-PLL) was utilized in this study. In this structure, the three-phase grid voltages are transformed into the dq reference frame using the estimated phase angle, and the instantaneous phase

angle is obtained by forcing the q-axis voltage component to zero.

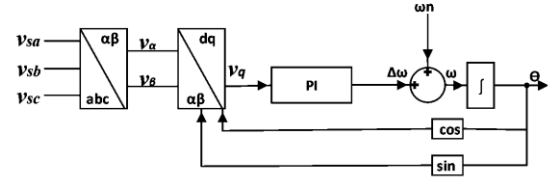


Figure 2. Block diagram of the Synchronous Reference Frame Phase-Locked Loop (SRF-PLL) for three-phase systems

The Park transformation is used to convert abc phase quantities into the dq frame. The general transformation relation is as follows:

$$\begin{bmatrix} u_d \\ u_q \\ u_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \sin(\omega t) & \sin(\omega t - \frac{2\pi}{3}) & \sin(\omega t + \frac{2\pi}{3}) \\ \cos(\omega t) & \cos(\omega t - \frac{2\pi}{3}) & \cos(\omega t + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix}$$

Here θ represents the instantaneous grid phase angle obtained by the PLL. Through this transformation, three-phase sinusoidal quantities can be expressed as stationary or slowly varying components in the dq frame.

In the SRF-PLL structure, the fundamental control objective is to satisfy the condition

$$v_q \rightarrow 0.$$

In an ideal synchronization state, the grid voltage vector aligns perfectly with the d-axis, and the q-axis voltage component approaches zero.

The phase angle obtained in this case is utilized both in the transformation of currents and voltages into the dq frame and during the inverse transformation stage.

In this study, the phase-to-neutral peak voltage was previously calculated as

$$V_{ph,peak} \approx 326.6 \text{ V}$$

The values of

$$V_d \approx 326.6 \text{ V}$$

and

$$V_q \approx -0.0525 \text{ V}$$

measured in the simulation demonstrate that the PLL has accurately aligned the grid voltage vector with the d-axis. The fact that the q-axis component is extremely small confirms that the synchronization error is at a negligible level.

As a result, the SRF-PLL structure has provided reliable phase tracking in the grid-connected inverter system used in this study and has established the necessary synchronous reference frame for the power and current control loops in the dq frame.

7. Active/Reactive Power Control and Derivation of dq Current References

In grid-connected inverter structures, active and reactive power control is performed via voltage and current components expressed in the dq frame. This approach, by aligning the grid voltage vector to the d-axis with the help of SRF-PLL, allows for the simplification of power equations and the decoupling of active/reactive power components.

The active and reactive power relations in the dq frame are as follows:

$$P = \frac{3}{2} (v_d i_d + v_q i_q)$$

$$Q = \frac{3}{2} (v_q i_d - v_d i_q)$$

Under ideal synchronization conditions, since $v_q \approx 0$ these expressions simplify to:

$$P \approx \frac{3}{2} v_d i_d$$

$$Q \approx -\frac{3}{2} v_d i_q$$

This demonstrates that active and reactive powers are primarily governed by the i_d and i_q currents, respectively.

In this study, the active power reference is set as:

$$P_{ref} = 25 \text{ kW}$$

and the reactive power reference is set as

$$Q_{ref} = 15 \text{ kVAr}$$

Using the simplified relations above, the current references in the dq frame are obtained from the power references as:

$$i_d^* = \frac{2 P_{ref}}{3 v_d}$$

$$i_q^* = -\frac{2 Q_{ref}}{3 v_d}$$

Since the d-axis voltage measured in steady-state is approximately

$$v_d \approx 326.6 \text{ V}$$

the theoretical current references are calculated as:

$$i_d^* = \frac{2 \cdot 25000}{3 \cdot 326.6} \approx 51.0 \text{ A}$$

$$i_q^* = -\frac{2 \cdot 15000}{3 \cdot 326.6} \approx -30.6 \text{ A}$$

The reference current values obtained in the simulation are approximately:

$$I_{d,ref} \approx 51.03 \text{ A}$$

$$I_{q,ref} \approx -30.63 \text{ A}$$

These results demonstrate that the current control loop successfully tracks the calculated references, maintaining the desired power flow between the inverter and the grid.

In this structure, current references are generated in the outer loop using active and reactive power references, while in the inner loop, these references are compared with the measured current components to form the inverter voltage references. Thus, a hierarchical control structure is established between power control and current regulation.

8. dq Current Control and SVPWM Modulation

In order to accurately transfer the i_d^* and i_q^* current references obtained from the outer power loop to the grid, current control in the dq frame is applied in the inner loop. In this structure, the measured i_d and i_q currents are compared with their reference values, and the error signals are processed by PI controllers to generate the inverter's voltage references in the dq frame.

The dq-axis voltage equations written over the grid-connected filter inductance are generally expressed as follows:

$$\begin{aligned} u_d &= L \frac{di_d}{dt} + Ri_d - \omega Li_q + e_d \\ u_q &= L \frac{di_q}{dt} + Ri_q + \omega Li_d + e_q \end{aligned}$$

Here u_d and u_q represent the inverter voltages in the dq frame, e_d and e_q represent the dq components of the grid voltage, L represents the filter inductance, and R represents the equivalent series resistance. The terms $-\omega Li_q$ and $+\omega Li_d$ in the equations represent the coupling between the d and q axes.

In current control, the error magnitudes are defined as:

$$\begin{aligned} e_{d,i} &= i_d^* - i_d \\ e_{q,i} &= i_q^* - i_q \end{aligned}$$

These error signals are processed by PI controllers to obtain the primary control terms. However, if the coupling between the dq axes and the effect of the grid voltage are not directly taken into account, transient performance may be limited. Therefore, feed-forward supported current control was used in this study.

In the feed-forward structure, the grid voltage components and cross-coupling terms are added to the controller output to generate the voltage references as follows:

$$\begin{aligned} u_d^* &= u_{d,PI} - \omega Li_q + e_d \\ u_q^* &= u_{q,PI} + \omega Li_d + e_q \end{aligned}$$

This structure aims to achieve faster tracking of current references and a more stable response to changes in power references.

The obtained u_d^* and u_q^* voltage references are transferred to the stationary frame via inverse transformation and applied to the SVPWM (Space Vector Pulse Width Modulation) block. SVPWM is a modulation method based on the representation of the reference voltage vector in the space vector plane for two-level three-phase inverters.

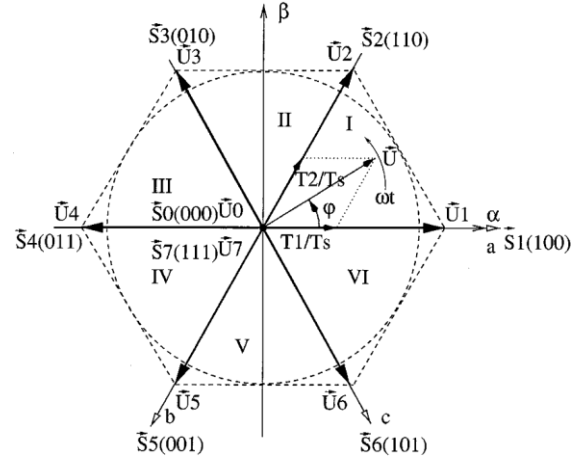


Figure 3. Switching vectors and sectors of Space Vector Pulse Width Modulation (SVPWM) for a three-phase inverter

In this method, the desired reference voltage vector is generated as an average by applying active and zero vectors for appropriate durations. One of the significant advantages of SVPWM is its more effective utilization of the DC bus voltage compared to the classical sinusoidal PWM method. This enables an increase in the fundamental component that can be produced within the linear modulation region. In practice, it is accepted that the maximum fundamental component obtained with SVPWM is approximately 1.15 times that of sinusoidal PWM. This advantage yields a voltage utilization improvement equivalent to that achieved with third-harmonic injection PWM.

This feature provides a critical advantage for this system, which connects to a 400 V phase-to-phase grid with a 700 V DC bus. Furthermore, SVPWM contributes to more balanced output voltage production through appropriate switching sequences and, when evaluated alongside the LCL filter, helps reduce the harmonic content of the current injected into the grid.

In this study, the SVPWM block generates the gate drive signals for the inverter's six switches using the voltage references obtained from the control loop. Consequently, the inverter is operated in a manner that can both track active and reactive power references and transfer low-harmonic current to the grid.

9. Simulation Results and Verification

The designed three-phase grid-connected inverter system has been verified through closed-loop simulation in the PSIM environment. In this section, active and reactive power tracking, the

accuracy of dq current references, PLL synchronization performance, phase currents, and the harmonic suppression effect of the LCL filter are evaluated.

9.1. Active and Reactive Power Tracking

In the simulation, the active power reference was applied as:

$$P_{ref} = 25 \text{ kW}$$

and the reactive power reference as:

$$Q_{ref} = 15 \text{ kVAr}$$

The power values measured in steady-state were obtained as:

$$\begin{aligned} P_{cal} &\approx 25.00 \text{ kW} \\ Q_{cal} &\approx 15.00 \text{ kVAr} \end{aligned}$$

These results indicate that the control structure accurately tracks the active and reactive power references.

9.2. Verification of dq Current References

Since the d-axis voltage measured in steady-state is:

$$V_d \approx 326.6 \text{ V}$$

the theoretical current references were calculated from the expressions as:

$$\begin{aligned} i_d^* &= \frac{2 P_{ref}}{3 V_d} \\ i_q^* &= -\frac{2 Q_{ref}}{3 V_d} \end{aligned}$$

and;

$$\begin{aligned} i_d^* &= \frac{2 \cdot 25000}{3 \cdot 326.6} \approx 51.0 \text{ A} \\ i_q^* &= -\frac{2 \cdot 15000}{3 \cdot 326.6} \approx -30.6 \text{ A} \end{aligned}$$

The reference current values obtained in the simulation are as follows:

$$\begin{aligned} I_{d,ref} &\approx 51.03 \text{ A} \\ I_{q,ref} &\approx -30.63 \text{ A} \end{aligned}$$

These results demonstrate that the dq current references are accurately generated from the power references.

9.3. PLL Synchronization Performance

The fundamental indicator of PLL performance is the α -axis voltage component being close to zero. In the simulation, the values were measured in steady-state as:

$$V_d \approx 326.6 \text{ V}$$

And

$$V_q \approx -0.0525 \text{ V}$$

The fact that the q-axis voltage component is extremely small demonstrates that the grid voltage vector has been successfully aligned with the d-axis and that phase synchronization has been accurately achieved.

9.4. Phase Currents and Power Level Verification

The RMS value of the phase currents before the filter was obtained as approximately

$$I_{a,rms} \approx I_{b,rms} \approx I_{c,rms} \approx 42.47 \text{ A}$$

The RMS value of the phase currents after the filter is approximately

$$I_{a,f,rms} \approx I_{b,f,rms} \approx I_{c,f,rms} \approx 42.13 \text{ A}$$

These values are consistent with the theoretical line current of approximately **42.1 A** obtained from the apparent power calculation. Furthermore, it was observed that the currents after the filter possess a smoother sinusoidal waveform.

9.5. Harmonic Suppression Effect of the LCL Filter

The impact of the LCL filter was evaluated based on Total Harmonic Distortion (THD) values. According to the simulation results, the pre-filter current THD was measured at approximately 1.746%, whereas the post-filter THD was reduced to 0.182%.

$$THD_{pre} \approx 1.746\%$$

$$THD_{post} \approx 0.182\%$$

Accordingly, the reduction rate in harmonic distortion is calculated using the following expression:

$$\eta_{THD} = \frac{THD_{pre} - THD_{post}}{THD_{pre}} \times 100$$

$$\eta_{THD} = \frac{1.746 - 0.182}{1.746} \times 100 \approx 89.6\%$$

This result demonstrates that the LCL filter effectively suppresses switching harmonics and significantly enhances the power quality of the current injected into the grid.

9.6. Summary of Simulation Results

Table 2. Key parameters obtained from the simulation

Parameter	Average / Measured Value	Definition
P_{cal}	25.00 kW	Active Power
Q_{cal}	15.00 kVAr	Reactive Power
$I_{d,ref}$	51.03 A	d-axis current reference
$I_{q,ref}$	-30.63 A	q-axis current reference
V_d	326.6 V	d-axis voltage component
V_q	-0.0525 V	q-axis voltage component
$I_{a,rms}$	42.47 A	Phase current before filter
$I_{a,f,rms}$	42.13 A	Phase current after filter

Parameter	Average / Measured Value	Definition
THD_{pre}	1.746 %	Current THD before filter
THD_{post}	0.182 %	Current THD after filter

In conclusion, the simulation findings verify the successful integrated operation of SRF-PLL based synchronization, active/reactive power control in the dq-frame, PI-based current regulation, the feed-forward structure, SVPWM modulation, and the LCL filter.

10. Conclusion and Evaluation

In this study, a 25 kW inverter system supplied by a 700 V DC bus and connected to a 400 V, 50 Hz three-phase grid was designed and analyzed under closed-loop control. The system architecture integrated SRF-PLL based synchronization, active/reactive power control in the dq-frame, PI-based current control with feed-forward support, LCL filtering, and SVPWM modulation.

Simulation results demonstrate that the active and reactive power references are accurately tracked, the PLL successfully achieves synchronization, and the post-filter currents exhibit a low-harmonic profile. In particular, the significant reduction in current THD achieved by the LCL filter indicates that the designed structure is well-suited for grid-connected inverter applications.

In conclusion, the established system has been found successful in terms of both power control and power quality, and the design objectives have been verified through simulation.

11. References

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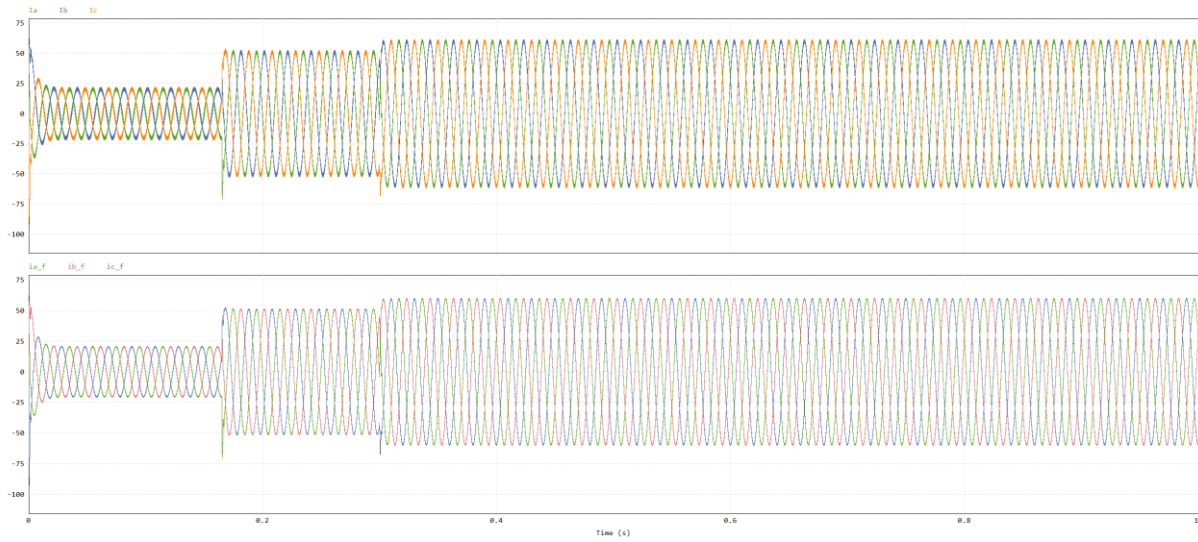


Figure 4: Current waveforms before filtering and current waveforms after filtering

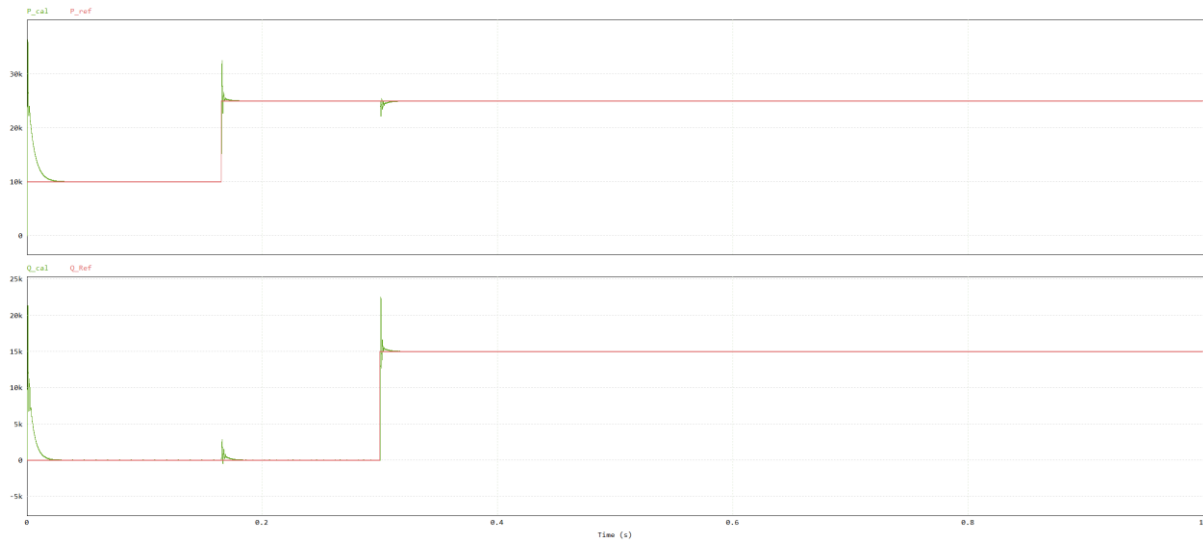


Figure 5: Active and Reactive Power Tracking

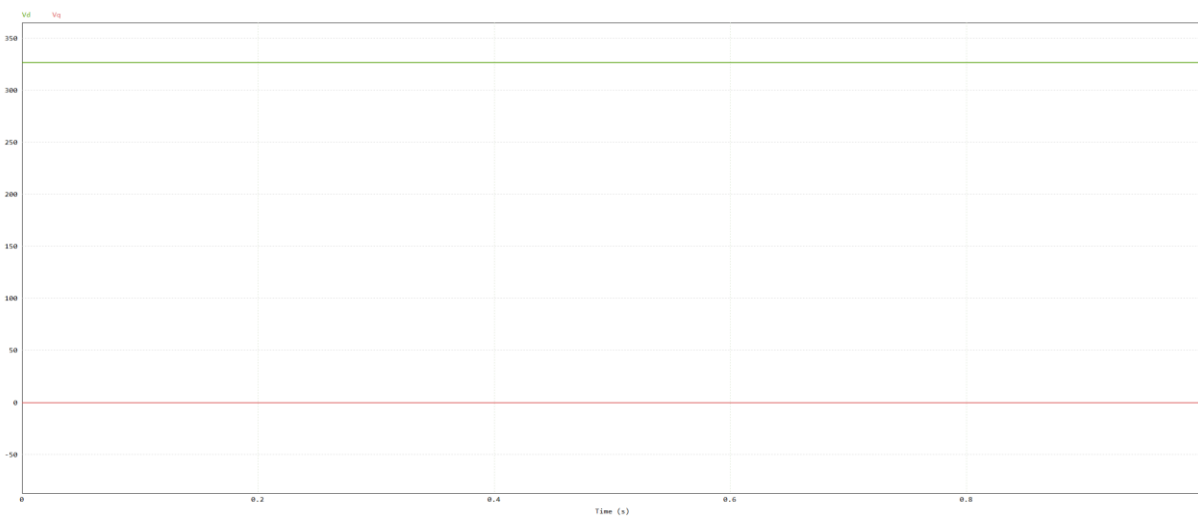


Figure 6: Vd and Vq waveforms

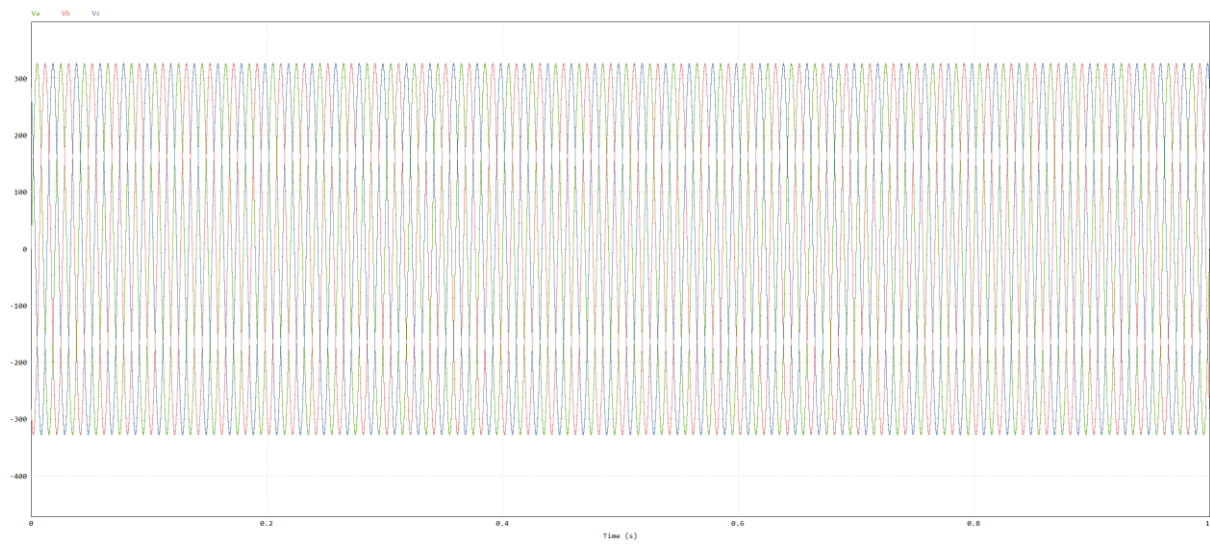


Figure 7: V_a , V_b , and V_c waveforms

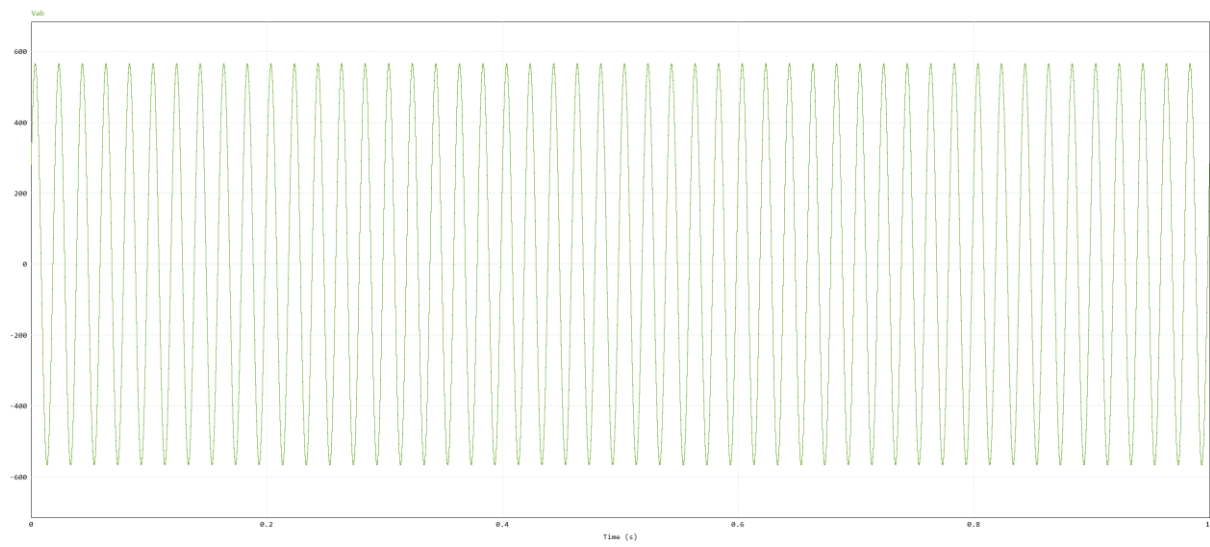


Figure 8: V_{ab} waveforms